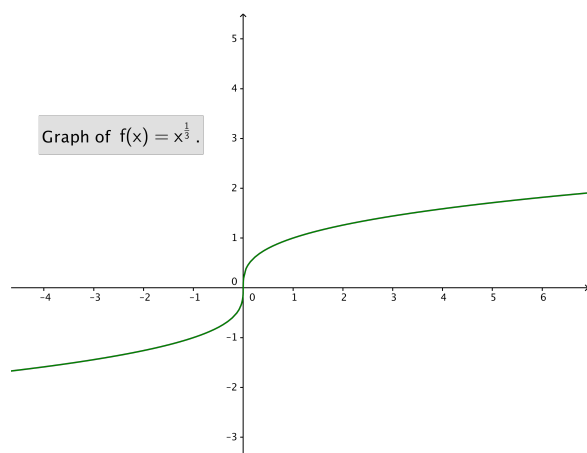


## Problem 4

## Problem 4

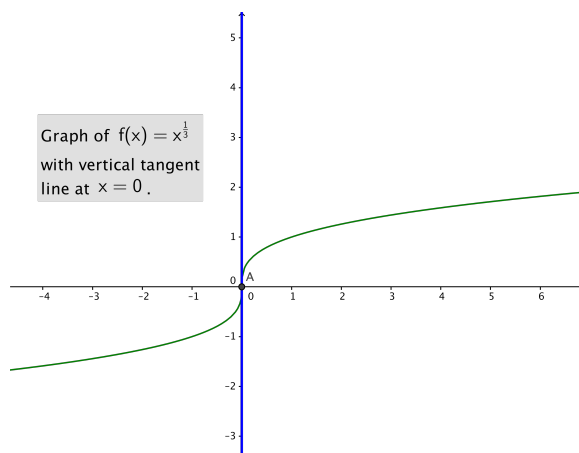
**Problem 1** Define the function  $f$  by  $f(x) = x^{1/3}$  and consider the graph of this function:



Which of the following two statements are true:

- (a) The graph of  $f$  has a tangent line at  $x = 0$ .

**Explanation.** This statement is **true**! The function  $f$  has a vertical tangent at  $x = 0$ :



(b) *The derivative  $f'(0)$  is defined.*

**Explanation.** This statement is **false!**

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} \frac{h^{1/3}}{h} \text{ which has form } \frac{0}{0} \\ &= \lim_{h \rightarrow 0^+} \frac{1}{h^{2/3}} \\ &= \infty \text{ because the limit is of the form: } \frac{\text{pos}}{0^+} \\ &\implies f'(0) \text{ is undefined} \end{aligned}$$